

Meet in the Middle

TASK

SUBMIT

RESULTS

STATISTICS

TESTS

Time limit: 1.00 s **Memory limit:** 512 MB

You are given an array of n numbers. In how many ways can you choose a subset of the numbers with sum x ?

Input

The first input line has two numbers n and x : the array size and the required sum.

The second line has n integers $t_1, t_2, \dots, t_n$: the numbers in the array.

Output

Print the number of ways you can create the sum x .

Constraints

- $1 \leq n \leq 40$
- $1 \leq x \leq 10^9$
- $1 \leq t_i \leq 10^9$

Example

Input:

4 5
1 2 3 2

Output:

3

$$n = 4$$

$$x = 5$$

$$\{2, 3\}$$

$$\{1, 2, 2\}$$

$$\{3, 2\}$$

3 ways

$$x = 5$$

$$\text{sumA} = \{0, 1, 2, 3\}$$

↑
i

$$\text{sumB} = \{0, 1, 2, 3, 4, 5, 6\}$$

↑ ↑

$$\begin{aligned}\text{remaining} &= x - 0 \\ &= 5 - 0 \\ &= 5\end{aligned}$$

$$\begin{aligned}\text{idx}_1 &= 3 \\ \text{idx}_2 &= 6 \\ \text{idx}_2 - \text{idx}_1 &= \\ 6 - 3 &= 3\end{aligned}$$

```
ll count = 0;
```

```
for (ll sum : sumA) {
```

```
    ll rem = x - sum;
```

```
    idx1 = lower-bound (sumB, rem)
```

```
    idx2 = upper-bound (sumB, rem)
```

```
    count = count + (idx2 - idx1)
```

```
}
```

```
return count;
```

Some Deep Analysis

Brute force Approach:

$$2^n \rightarrow 2^{40} \approx 10^{12} \text{ operation.}$$

$$10^8 \text{ operation} \longrightarrow 1 \text{ sec}$$

$$1 \quad , \quad \longrightarrow \frac{1}{10^8} \text{ sec}$$

$$10^{12} \quad , \quad \longrightarrow \frac{1}{10^8} \times 10^{12}$$

$$\approx 10^4$$

Meet in the middle approach: